Lithium-ion Battery State of Health (SOH) Estimation Using WOA-CNN-LSTM

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Abstract

Accurate State of Health (SOH) estimation is essential for the safe and efficient operation of lithium-ion batteries in electric vehicles and energy storage systems. This project presents a hybrid deep learning approach for battery SOH prediction using the NASA dataset (B0005, B0006, B0007). Four health factors are extracted from charge and discharge cycles, followed by robust outlier detection using Local Outlier Factor (LOF) and data cleaning via linear interpolation. Three models are implemented and compared: a baseline Plain LSTM, a standard CNN-LSTM with fixed hyperparameters, and a WOA-optimized CNN-LSTM model. Results demonstrate that the WOA-CNN-LSTM model achieves superior prediction accuracy, highlighting the benefits of hyperparameter optimization and hybrid architecture for battery management systems.

1 Introduction

Lithium-ion batteries have become the dominant energy storage technology in electric vehicles, renewable energy systems, and portable electronics due to their high energy density, long cycle life, and relatively low self-discharge rate. However, battery performance degrades over time due to complex electrochemical aging mechanisms, leading to capacity fade and increased internal resistance. Accurate estimation of the battery State of Health (SOH), defined as the ratio of current capacity to nominal capacity, is essential for ensuring operational safety, optimizing charging strategies, and predicting remaining useful life (RUL) [1, 2].

Traditional SOH estimation approaches based on electrochemical models or direct capacity measurements are often impractical for real-time applications, as they require intrusive testing procedures or specialized laboratory equipment [3]. In contrast, data-driven methods have gained significant attention due to their ability to learn complex degradation patterns directly from measurable signals such as voltage, current, and temperature [4, 5]. Among these, deep learning techniques have demonstrated superior performance in capturing nonlinear and temporal dependencies inherent in battery degradation processes.

This project focuses on the reproduction and enhancement of a hybrid CNN-LSTM architecture optimized using the Whale Optimization Algorithm (WOA) for battery SOH estimation. Using the NASA lithium-ion battery dataset, four health factors are extracted from charge and discharge cycles, followed by systematic outlier detection and data cleaning. Three models are then evaluated: a baseline LSTM, a CNN-LSTM with fixed hyperparameters, and a WOA-optimized CNN-LSTM. The objective is to demonstrate improved prediction accuracy through hyperparameter optimization while maintaining computational efficiency suitable for real-time battery management systems (BMS).

The complete implementation of the proposed WOA-CNN-LSTM framework is publicly available on GitHub at: `https://github.com/EllieSalimi96/ENSF617`.

2 Related Works

Early studies on battery SOH estimation primarily relied on equivalent circuit models (ECMs) and electrochemical impedance spectroscopy (EIS) [3]. With the emergence of machine learning, methods such as Support Vector Regression (SVR) and Gaussian Process Regression (GPR) were introduced, offering improved flexibility but limited capability in modeling long-term temporal dependencies [2, 6].

Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) networks, have been widely adopted due to their effectiveness in modeling sequential degradation data [1]. Hybrid architectures combining convolutional layers with LSTM have further improved performance by capturing both local feature patterns and temporal dynamics [7, 8]. Additionally, metaheuristic optimization techniques such as Particle Swarm Optimization (PSO) and Whale Optimization Algorithm (WOA) have been employed to enhance model performance through automated hyperparameter tuning [9, 10].

More recently, attention-based mechanisms and transformer-inspired models have been introduced to better capture long-range dependencies and improve interpretability in SOH estimation tasks [11–13]. Despite these advancements, many existing studies suffer from limitations such as small-scale datasets, insufficient handling of noisy measurements, or lack of consistent benchmarking across different model architectures. This work addresses these challenges by performing a systematic comparison of baseline and optimized deep learning models on a cleaned and standardized NASA dataset, with explicit evaluation of preprocessing and optimization effects.

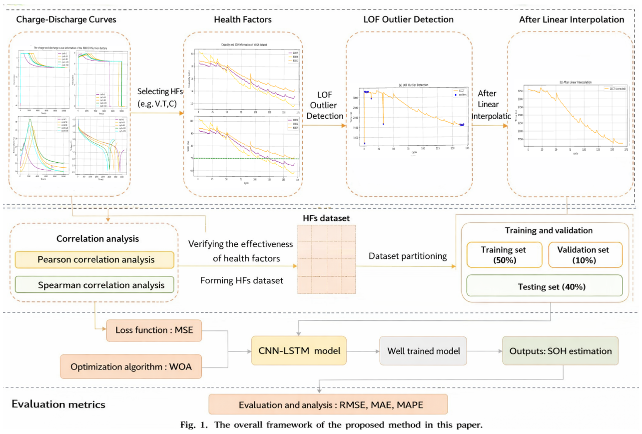

Figure 1: The overall framework of the proposed method in this project.

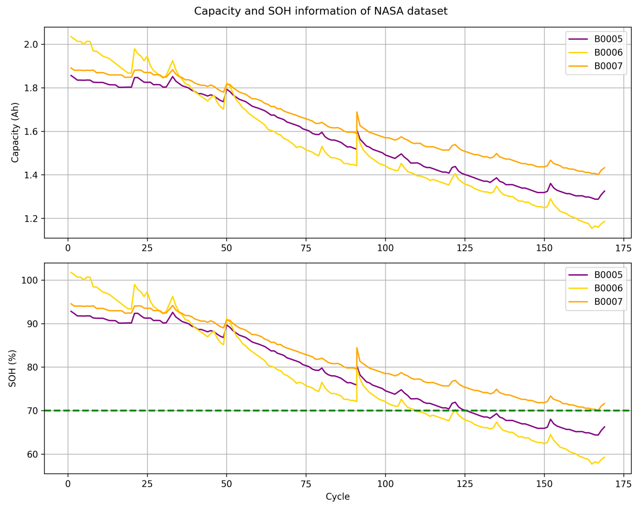

Figure 2: Capacity and SOH information of NASA dataset (B0005, B0006, B0007).

3 Methodology

Current methods for State of Health (SOH) estimation of lithium-ion batteries often suffer from limited accuracy, poor generalization across different batteries, and insufficient capability to capture complex temporal dependencies in degradation patterns. To address these challenges, this study proposes a hybrid deep learning framework based on CNN combined with Long Short-Term Memory (CNN-LSTM) architecture, further enhanced by the Whale Optimization Algorithm (WOA) for hyperparameter tuning. The CNN layers effectively extract local features from health factor sequences, while the LSTM layers capture long-term temporal dependencies in battery degradation patterns. This combination addresses the limitations of traditional methods and improves overall SOH estimation performance.

Figure 1 illustrates the overall framework of the proposed methodology. Four health factors (HF1–HF4) are extracted from raw charge-discharge data and preprocessed using LOF detection and linear interpolation. The cleaned dataset is split into training, validation, and testing sets. Finally, three proposed models are trained and evaluated to estimate battery SOH.

3.1 Datasets

The dataset used for this work is the publicaly available lithium-ion dataset at the National Aeronautics and Space Administration (NASA) research center. The selected groups of lithium-ion batteries, B0005, B0006, and B0007, are rated at 2 Ah. These batteries have 168 charge–discharge cycles. Due to the aging mechanisms of lithium-ion batteries, if the actual capacity of the lithium-ion battery in a device decreases to 70% of its initial rated capacity (e.g., from 2.0 Ah to 1.4 Ah), it is considered to have reached failure and should be replaced immediately. The State of Health (SOH) is defined in terms of capacity as SOH $= \frac{C_{\text{cur}}}{C_{\text{init}}} \times 100\%$, where C_{cur} is the current actual capacity and C_{init} is the initial rated capacity of the battery.

Figure 2 shows that all batteries used in this study exhibit a clear decreasing trend in capacity and SOH over cycles, confirming the expected aging behavior.

3.2 Data Preprocessing

Due to the complex aging mechanisms and discharge uncertainty in lithium-ion batteries, their State of Health (SOH) is influenced by multiple factors such as current, voltage, and temperature. In this study, both the constant current-constant voltage charging and constant current discharging processes were analyzed. Accordingly, four capacity-related health factors (HF1: CCCT, HF2: time to peak temperature during charging, HF3: time to peak charging voltage, and HF4: discharge time to cutoff voltage) were extracted as input features for the SOH estimation models.

The charge–discharge curves of battery B0005 at different cycle counts are shown in Fig. 3.

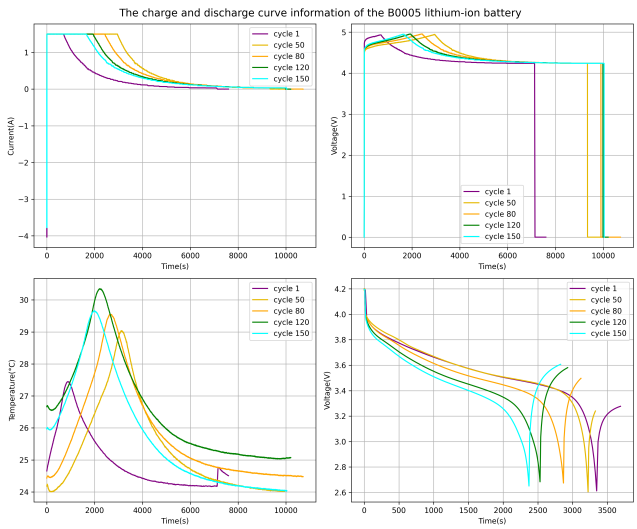

Figure 3: Charge and discharge curve information of the B0005 lithium-ion battery.

Local Outlier Factor(LOF) method is used to find the anomalies including zero-value anomalies and significant deviations from neighboring samples throughout the dataset. After identifying outliers in the dataset using the LOF method, linear interpolation is applied to restore data continuity and integrity. This technique estimates missing values

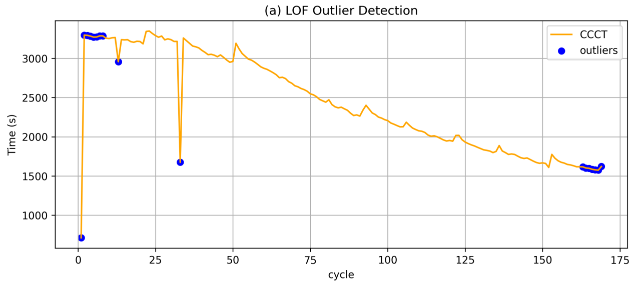

Figure 4: LOF outlier detection for CCCT of battery B0005.

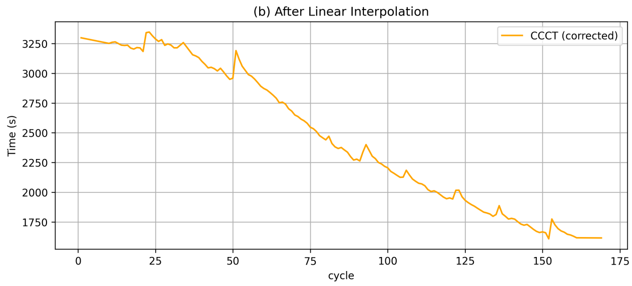

Figure 5: CCCT values after linear interpolation for battery B0005.

by connecting neighboring normal data points identified by LOF with a straight line. By leveraging information from surrounding normal points, linear interpolation effectively fills in the outlier values while preserving the overall trend of the health factor sequence.

3.3 Dataset Construction

After preprocessing, an effective health factors (HFs) dataset is obtained. To verify their effectiveness, Pearson (PCC) and Spearman (RHO) correlation analyses are performed. The Pearson correlation is given by $PCC = \frac{\sum(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum(x_i-\bar{x})^2\sum(y_i-\bar{y})^2}}$, while the Spearman correlation is defined as $RHO = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$. These complementary analyses provide a comprehensive evaluation of the selected HFs.

Table 1 presents the Pearson (PCC) and Spearman (RHO) correlation coefficients between the four health factors and battery capacity. All health factors show strong positive correlations, validating their suitability as input features for SOH estimation.

Table 1: Pearson and Spearman correlation results between health factors and capacity

HFs	B0005		B0006		B0007	
	PCC	RHO	PCC	RHO	PCC	RHO
HF1	0.9973	0.9931	0.9940	0.9948	0.9959	0.9905
HF2	0.9962	0.9939	0.9681	0.9845	0.9689	0.9848
HF3	0.6845	0.6624	0.7675	0.6434	0.6795	0.6838
HF4	0.9997	0.9975	0.9985	0.9986	0.9989	0.9977

3.4 Model Architecture

Three model architectures were compared in this study: a baseline Plain LSTM, a standard CNN-LSTM with fixed hyperparameters, and a WOA-optimized CNN-LSTM. For each battery, the cleaned data was split into 50% training, 10% validation, and 40% testing sets. All models were trained in a supervised fashion for 200 epochs using the Adam optimizer and Mean Squared Error (MSE) loss function. The Plain LSTM consists of a single LSTM layer with 128 hidden units followed by a fully connected output layer. The CNN-LSTM model integrates a 1D convolutional layer (with 32 output channels and kernel size 3) for local feature extraction before feeding the features into the LSTM layer. The WOA-CNN-LSTM employs the same hybrid CNN-LSTM architecture, but its key hyperparameters—including learning rate, LSTM hidden size, batch size, and dropout rate—are automatically optimized using the Whale Optimization Algorithm (WOA). All models take sliding windows of length 10 from the preprocessed health factors as input and predict the battery capacity at the end of the window.

4 Results and Discussion

RMSE, MAE, and MAPE were selected as evaluation metrics because they are widely adopted in battery SOH estimation literature, enabling direct and fair comparison with previous studies. While RMSE and MAE quantify the absolute prediction error in capacity (Ah), MAPE provides an intuitive percentage-based measure of relative accuracy that is particularly meaningful for interpreting SOH degradation.

Table 2: Performance comparison of the three models in terms of RMSE and MAPE (SOH %)

Model	B0005	B0006	B0007
RMSE (%)			
Plain LSTM	14.06	13.29	9.55
CNN-LSTM	10.517	7.936	6.420
WOA-CNN-LSTM	**0.775**	**1.950**	**3.842**
MAPE (%)			
Plain LSTM	13.54	9.45	8.90
CNN-LSTM	5.350	6.385	4.899
WOA-CNN-LSTM	**0.513**	**1.538**	**3.792**

Table 3: Performance comparison of the three models in terms of MAE (SOH %).

Model	B0005	B0006	B0007
Plain LSTM	20.26	15.37	12.34
CNN-LSTM	5.36	10.36	6.77
WOA-CNN-LSTM	**0.77**	**2.51**	**5.22**

The performance of the three models is summarized in Tables 2 and 3. The WOA-CNN-LSTM model consistently achieves the best results across all batteries, significantly reducing both RMSE and MAE compared to the Plain LSTM and standard CNN-LSTM. For B0005, it attains an excellent MAE of 0.513% (over 90% improvement over Plain LSTM). Although absolute errors are higher on B0007 due to more complex degradation, the WOA-CNN-LSTM still provides the lowest RMSE, MAPE, and MAE on every battery.

Table 4 shows the best hyperparameters found by the Whale Optimization Algorithm (WOA) for each battery, along with the final performance metrics. The WOA algorithm successfully identified battery-specific optimal configurations.

Table 4: WOA optimized hyperparameters and final performance

Parameter	**B0005**	**B0006**	**B0007**
Learning Rate	0.00828	0.00825	0.00831
LSTM Hidden Size	373	468	415
Batch Size	45	58	27
Dropout Rate	0.055	0.015	0.047
Final MAE (%)	0.513	1.538	3.792
Final RMSE (%)	0.775	1.950	3.842
Final MAPE (%)	0.77	2.51	5.22

Table 4 presents the optimal hyperparameters found by the Whale Optimization Algorithm (WOA) for each battery and the final performance of the WOA-CNN-LSTM model.

Figure 6 shows the prediction performance of the WOA-CNN-LSTM model against the ground truth SOH for the three batteries. The model accurately captures the overall degradation trend, with minimal deviation on B0005 and B0006. On B0007, slightly larger deviations appear in later cycles, consistent with its higher error metrics. Figure 7 compares the prediction performance of the Plain LSTM and WOA-CNN-LSTM models against the ground truth SOH. The WOA-CNN-LSTM model tracks the true degradation trend with significantly smaller deviations than the Plain LSTM, visually demonstrating the clear improvement achieved through CNN feature extraction and WOA hyperparameter optimization.

Strengths of the proposed methodology:

- Effective data preprocessing using LOF outlier detection and linear interpolation significantly improves data quality and model stability.
- The WOA automatically identifies near-optimal, battery-specific hyperparameters.
- The hybrid CNN-LSTM architecture successfully captures both local features from health factors and long-term temporal dependencies in battery degradation.

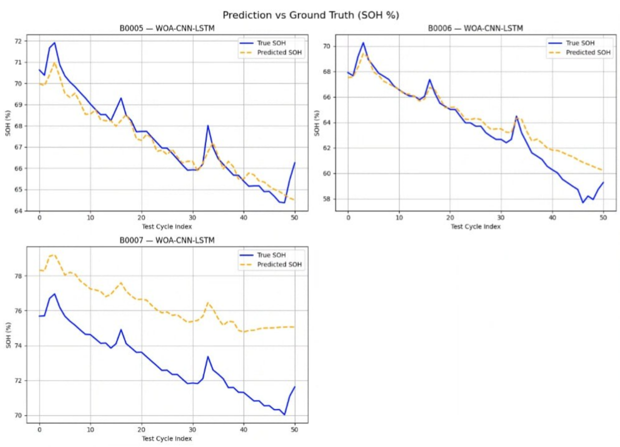

Figure 6: Prediction vs Ground Truth (SOH %) for the WOA-CNN-LSTM model.

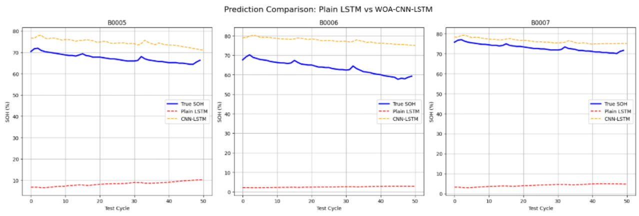

Figure 7: Prediction Comparison: Plain LSTM vs WOA-CNN-LSTM.

Limitations:

- The study was limited to only three batteries from the NASA dataset, reducing the generalizability of the results.
- The model was trained and evaluated exclusively on laboratory-controlled data and has not been tested under real-world operating conditions.

5 Conclusions

This study developed a hybrid CNN-LSTM model optimized by the Whale Optimization Algorithm for accurate lithium-ion battery State of Health estimation. The proposed WOA-CNN-LSTM framework demonstrated superior prediction performance compared to both the baseline Plain LSTM and the standard CNN-LSTM across the NASA dataset.

The results confirm that combining convolutional feature extraction, LSTM temporal modeling, and metaheuristic hyperparameter optimization can significantly improve SOH prediction accuracy. While the current work was limited to laboratory data and three batteries, it provides a practical and well-documented methodology for battery health monitoring.

Future work will focus on incorporating additional datasets, exploring advanced attention mechanisms, and validating the model under real-world operating conditions for eventual deployment in battery management systems. Overall, this research contributes toward more reliable and efficient lithium-ion battery operation in electric vehicles and energy storage applications.

6 AI Aknowledgment

The author acknowledges the use of Grok, an AI assistant, for assistance in debugging the Python code and for editing and refining the structure and language of this report. All code implementations, analyses, results, and the final content of this work are the original contributions of the author.

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